

Real-Time Computing for Particle Physics: From AI on FPGA to AI Agents

**DATE****Friday, 9/4****TIME****12:30-1:30 PM****LOCATION****AgSci 220****SPEAKER**

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ABSTRACT

Modern particle physics experiments are among the world's most demanding computing systems, processing enormous data streams in real time. Experiments such as Belle II require detector information to be analyzed within only a few microseconds for event triggering and data acquisition. This seminar introduces the computing challenges behind modern collider experiments and presents recent developments in FPGA-based real-time computing, AI acceleration on hardware, and AI-agent systems for chip design and scientific data analysis. The talk will also highlight opportunities for interdisciplinary collaboration between computer science and experimental physics through research and education at UH.

RESEARCH THEMES**REAL-TIME SYSTEMS****FPGA ACCELERATION****AI AGENTS****SCIENTIFIC DATA**